

Systematic Review on Measurement and Visualization of Cryptocurrency Adoption, Network Activity, and Financial Performance Using On-Chain and Market Data

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1 Introduction

The cryptocurrency ecosystem has introduced a fundamental transformation in digital finance through its decentralized architecture, transparency, and publicly verifiable ledger of economic activity. Unlike traditional financial systems, where transactional data are often proprietary, delayed, or fragmented, blockchain networks generate continuous and openly accessible records of user behavior, transaction activity, and network security. This data-rich environment enables real-time observation of adoption patterns, network utilization, and economic flows, making on-chain data analysis a central approach for evaluating the underlying health, valuation, and long-term viability of digital assets. As cryptocurrencies increasingly move beyond speculative use toward broader institutional adoption and heightened regulatory attention, understanding how network-level activity relates to financial performance has become both timely and necessary. The practical relevance of this research domain is substantial. Investors and market analysts require reliable indicators to assess value and manage risk in markets known for extreme volatility. Protocol developers depend on

adoption and network metrics to improve scalability, security, and incentive mechanisms. At the same time, policymakers and regulators increasingly rely on empirical evidence to design informed governance and oversight frameworks. From a scientific standpoint, the convergence of blockchain technology, network science, and financial econometrics offers a unique empirical setting for examining theories of market efficiency, diffusion and adoption dynamics, and the valuation of decentralized systems. Empirical research in this area typically integrates two complementary data streams. The first consists of on-chain data obtained from blockchain explorers and analytics platforms, capturing measures such as daily active addresses, new wallet creation, transaction count, transaction volume, network fees, and consensus participation through hash rate or staking activity. These metrics reflect the actual usage, security, and operational state of a blockchain network. The second stream comprises market data sourced from exchanges and data aggregators, including price returns, volatility, market capitalization, and liquidity, which capture market valuation and investor behavior. Analytical workflows commonly employ quantitative methods such as time-series econometrics, regression analysis, network analysis, and machine learning techniques. These methods are often used to construct derived economic indicators—most notably the Network Value to Transactions (NVT) ratio—which aim to link on-chain usage with market valuation.

Despite growing academic and industry interest, several core technical challenges continue to limit consensus in the literature. On-chain data are inherently heterogeneous and noisy, influenced by factors such as address reuse, self-transactions, and exchange-related activity that may not correspond to genuine economic use. Cryptocurrency markets themselves exhibit high volatility and non-stationarity, complicating statistical inference and model stability. Moreover, most studies rely on observational data, making causal relationships difficult to establish. As a result, there is little agreement on which on-chain metrics are most informative for forecasting returns, explaining volatility, or assessing fundamental value. Existing research remains fragmented, with many studies focusing on single cryptocurrencies, short time horizons, or examining network and market indicators in isolation. To address these limitations, this paper presents a Systematic Literature Review (SLR) that consolidates and critically evaluates empirical research on the measurement and visualization of cryptocurrency adoption, network activity, and financial performance. Guided by a conceptual framework in which user adoption drives network activity, which in turn influences market outcomes, the review is structured around the following research questions:

- **RQ1:** Do increases in new wallet creation and active address counts lead to higher transaction activity and measurable market growth?
- **RQ2:** Which on-chain metrics most effectively predict cryptocurrency returns and market volatility?
- **RQ3:** How does the relationship between transaction count and transaction volume distinguish genuine network usage from speculative behavior?
- **RQ4:** How do transaction fee dynamics and consensus participation levels (hash rate or staking) influence network adoption, security, and long-term asset value?

- **RQ5:** Do market capitalization and liquidity ratios serve as reliable signals of major cryptocurrency market cycle shifts?
- **RQ6:** Does the Network Value to Transactions (NVT) ratio accurately identify overvalued or undervalued assets, and does its predictive performance vary across different cryptocurrency types?

By systematically synthesizing evidence from 50 selected studies, this review aims to identify robust indicators, clarify methodological best practices, and highlight critical gaps for future research. The remainder of the manuscript is structured as follows: Section 2 presents the research questions guiding the review; Section 3 describes the PRISMA 2020–based methodology; Section 4 reports the synthesized findings mapped to the research questions; Section 5 discusses implications, limitations, and future research directions; and Section 6 concludes the paper.

2 Literature Review

Research on cryptocurrencies has evolved rapidly, moving from early studies of Bitcoin’s monetary properties to more advanced, data-driven analyses of market behavior, network dynamics, and valuation. This shift has been enabled by the growing availability of detailed on-chain data and the use of advanced econometric and computational methods. Despite this progress, existing review studies often focus on isolated areas, such as investment performance, technological foundations, or forensic analysis, resulting in a fragmented understanding of the field. In particular, a focused synthesis that examines how on-chain adoption and activity metrics are measured and linked to financial performance remains limited. This section reviews key empirical findings from primary studies and situates the present work within the broader landscape of existing surveys.

2.1 Insights From Research Articles

The empirical literature reveals several consistent findings, methodological trends, and limitations, which are discussed according to major metric categories.

Adoption and User Activity Metrics

User adoption is widely regarded as a core determinant of cryptocurrency network growth. Several empirical studies use new wallet creation and active address counts as proxies for adoption and find a positive relationship with market capitalization and transaction activity [1], [2]. Evidence from Bitcoin and Ethereum markets suggests that sustained increases in active addresses precede periods of higher network usage and valuation [3]. Research indicates that active token dynamics are crucial for understanding fundamental adoption and subsequent valuation [50]. Beyond pure network effects, behavioral factors also play a significant role in adoption. Studies show that trust in blockchain technology and financial literacy significantly influence user participation [4], [5]. Social influence and investor sentiment further amplify adoption dynamics, particularly in emerging markets [6]. Cross-country analyses indicate

that adoption is shaped by both domestic macroeconomic conditions and international spillover effects, highlighting the global nature of cryptocurrency diffusion [7], [8].

Network Activity and Transaction Behavior

Transaction-level metrics, including transaction counts and transferred volume, are commonly employed to measure network activity. Empirical findings suggest that higher transaction activity contains predictive information about future price movements and market liquidity [9], [10]. However, measurement challenges remain. Research on UTXO-based blockchains demonstrates that raw transaction volumes may overstate actual economic activity due to protocol-generated change outputs [11]. Transaction fees are another important indicator of network demand. Queueing-based models show that fees increase non-linearly with congestion, reflecting competition for block space [12], [13]. Fee dynamics are also closely linked to network security, as they provide incentives for miners and validators to maintain network integrity [14].

On-Chain Metrics and Financial Performance

A growing literature investigates the relationship between on-chain activity and financial performance indicators such as returns and volatility. Traditional econometric studies find statistically significant links between transaction activity and returns across major cryptocurrencies [15], [16]. More recent research demonstrates that machine learning models outperform linear approaches by capturing non-linear dependencies among on-chain variables [17], [18]. Deep learning architectures, including LSTM and hybrid models, have proven particularly effective for volatility forecasting in cryptocurrency markets [19], [20]. Adaptive and dynamically tuned models further enhance predictive accuracy and generate higher economic value compared to static parameter specifications [21]. High-frequency analyses reveal that on-chain indicators also contribute to cross-cryptocurrency return predictability and liquidity transmission [22], [23].

Valuation Metrics and Derived Indicators

Derived valuation metrics play an important role in assessing cryptocurrency fundamentals. The Network Value to Transactions (NVT) ratio is widely applied as a measure of relative valuation and systemic risk [24]. Empirical evidence suggests that elevated NVT values are associated with overvaluation and heightened market instability [25]. Recent studies propose refinements to the NVT ratio by adjusting for transaction inflation and token dormancy, which significantly improves its reliability [26]. Connectedness analyses based on NVT dynamics reveal heterogeneous spillover effects across cryptocurrencies, with Proof-of-Stake assets exhibiting lower systemic interconnectedness [27].

Consensus Mechanisms, Security, and Sustainability

Consensus mechanisms influence both economic incentives and network security. Empirical research confirms a strong relationship between hash rate and security in Proof-of-Work systems [28]. In Proof-of-Stake networks, simulation-based studies

highlight the risk of wealth concentration and propose mechanisms to promote decentralization [29]. Energy consumption remains a critical concern in blockchain research. Studies evaluating mining efficiency show that cryptocurrency mining can be both more and less carbon-efficient than traditional industries, depending on energy sources [30]. Alternative consensus designs, such as energy-efficient Proof-of-Work variants, aim to reduce environmental impact while maintaining security properties [31].

Cross-Market Interactions and Macroeconomic Effects

Cryptocurrency markets are increasingly integrated with traditional financial systems. Empirical evidence indicates significant volatility spillovers among major cryptocurrencies [32] and between cryptocurrencies and conventional asset classes [33]. Advanced network-based and graph learning approaches further capture complex inter-market dependencies [34]. Macroeconomic fundamentals also influence cryptocurrency returns. Quantile regression studies show that variables such as interest rates, exchange rates, and production indices exert asymmetric effects across market conditions [35]. Uncertainty indices and macro-financial shocks affect returns differently across time horizons, as demonstrated by wavelet-based analyses [36].

Data Quality and Methodological Innovations

Reliable analysis of on-chain data requires careful preprocessing and validation. Large-scale entity-clustered transaction datasets have been developed to improve measurement accuracy and reproducibility [37], [38]. Network mining techniques, such as temporal motif analysis, uncover recurring behavioral patterns in transaction networks [39]. Methodological advances also extend to applied domains. Graph neural networks and transformer-based models have been successfully applied to detect illicit transactions in cryptocurrency systems [40], [41]. Optimization-based approaches further improve transaction efficiency and energy usage in blockchain networks [42]. Trading-oriented studies evaluate the usefulness of blockchain metrics as signals for investment strategies [43], [44].

Conceptual Perspectives and Reviews

Conceptual studies critique price-centric narratives of cryptocurrencies and emphasize their broader economic and social value [45]. Sector-specific applications, such as blockchain-based fan tokens, illustrate the expanding use cases of distributed ledger technologies [46]. Existing systematic reviews organize cryptocurrency research into thematic clusters but often treat on-chain metrics as secondary elements [47], [48]. Behavioral-focused reviews synthesize adoption drivers using established technology acceptance frameworks [49].

2.2 Findings From Systematic Reviews and Surveys

This study differs from prior reviews by explicitly focusing on empirical measurement of on-chain adoption and activity metrics and their link to financial performance. It integrates evidence across methodological approaches, combining econometric and machine learning perspectives into a unified framework. By synthesizing data quality

considerations, predictive modeling innovations, and valuation approaches, including insights from active token valuation frameworks [50], the review provides a coherent understanding of how on-chain behavior relates to cryptocurrency market dynamics. It highlights gaps in existing research, identifies avenues for future investigation, and emphasizes the practical relevance of network activity indicators for both researchers and practitioners.

3 Methodology

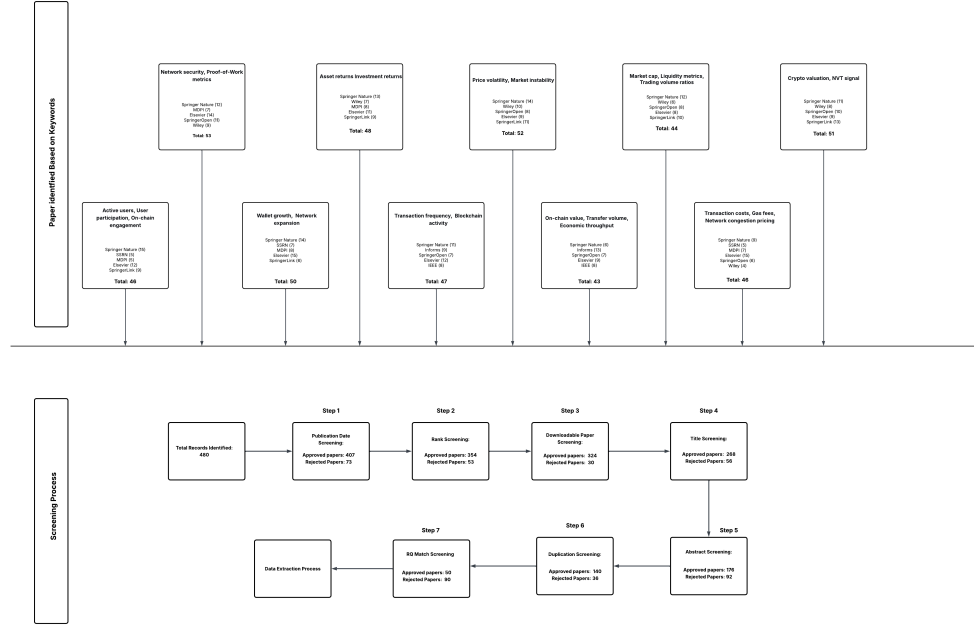


Fig. 1: Systematic Screening Processes For Paper Selection

This systematic literature review (SLR) followed the PRISMA 2020 framework to ensure a transparent, reproducible, and bias-controlled study selection process. The objective was to systematically analyze empirical studies that investigate ten predefined cryptocurrency on-chain and market indicators from a data analytics perspective: Hash Rate/Consensus Participation, Price Returns, Volatility Measures, Market Capitalization and Liquidity Ratios, Network Value to Transactions (NVT) ratio, Daily Active Addresses, New Wallet Creation, Transaction Count, Transaction Volume, and Network Fees.

3.1 Search Strategy and Information Sources

The search strategy was built from explicit keywords directly representing the targeted indicators and their analytical context.

Primary keywords:

network security, Proof-of-Work metrics, asset returns, investment returns, price volatility, market cap, liquidity metrics, trading volume ratios, crypto valuation, NVT signal, wallet growth, transaction costs, transaction frequency, on-chain value

Contextual keywords:

blockchain analytics, statistical modeling, econometric analysis, machine learning, time series forecasting, market volatility, network activity. These terms were combined using Boolean operators (AND, OR) to improve recall and precision. Searches were conducted in Google Scholar, IEEE Xplore, ScienceDirect, Springer-Link/SpringerOpen, Wiley Online Library, INFORMS, and SSRN. Forward and backward citation snowballing was also applied. Only English-language papers published between 2021 and 2026 were considered. The search process yielded 480 initial records.

3.2 Eligibility Criteria

Eligibility logic ensured inclusion of high-quality empirical studies directly analyzing at least one of the targeted indicators using data analytics methods and reporting measurable outcomes related to cryptocurrency market or network behavior. Full-text availability was mandatory.

3.2.1 Inclusion Criteria

- Publication year between 2021 and 2026
- English language
- SCImago Q1 journal or CORE A/A* conference
- Empirical analytical/statistical/ML-based study
- Direct analysis of at least one targeted indicator
- Measurable reported outcomes
- Downloadable full text

3.2.2 Exclusion Criteria

- Reviews, editorials, or conceptual papers without experiments
- Unrelated to cryptocurrency or blockchain analytics
- Published in low-rank venues
- Non-English papers
- No measurable indicator-related outcomes
- Inaccessible full text

3.3 Screening Processes

A structured multi-stage screening pipeline was applied to systematically reduce the initially identified 480 records to the final 50 studies included for data extraction. The screening process followed a PRISMA-aligned workflow and was implemented sequentially according to predefined steps. Zotero was used for reference management, duplicate detection, and metadata comparison, while spreadsheet-based logs were maintained to track approval and rejection counts at each screening stage. The initial search across all selected databases and digital libraries resulted in 480 records after aggregation of results from all keyword–database combinations.

3.3.1 Duplication Screening

Duplicate records were identified using DOI matching, metadata comparison, and fuzzy title matching within Zotero. When multiple versions of the same study were detected, the final published journal or conference version was retained. This screening step resulted in 140 approved papers, with 36 papers removed as duplicates.

3.3.2 Title Screening

Titles were examined to assess relevance to cryptocurrency on-chain metrics, market indicators, and empirical data analytics methods. Papers clearly unrelated to the research scope were removed at this stage. During title screening, 268 papers were approved, and 56 papers were rejected.

3.3.3 Publication Date Screening

In the first screening step, records were filtered based on publication year to retain only studies published within the defined time window of 2021–2026. Papers published outside this range were excluded to ensure methodological recency and relevance. At this stage, 407 papers were approved, while 73 papers were rejected due to being outside the specified publication period.

3.3.4 Abstract Screening

Abstracts of the remaining papers were reviewed to confirm empirical methodology, indicator coverage, and the presence of measurable outcomes aligned with the research objectives. This stage resulted in 176 papers being approved, while 92 papers were rejected for lacking sufficient methodological or topical relevance.

3.3.5 Rank Screening

The remaining studies were evaluated based on venue quality using SCImago Journal Rank (Q1) and CORE Conference Rank (A/A)* criteria. This step ensured that only high-quality, peer-reviewed research progressed further in the review process. Following rank screening, 354 papers were approved, and 53 papers were rejected for not meeting the required venue quality thresholds.

3.3.6 RQ Match Screening

In the final screening step, each study was mapped to the specific research questions associated with the predefined cryptocurrency indicators. Only studies providing direct empirical evidence relevant to at least one research question were retained. Following RQ match screening, 50 papers were approved, and 90 papers were rejected, resulting in the final corpus used for data extraction.

3.4 Data Extraction

A standardized data extraction form was designed, pilot-tested, and then applied uniformly to the final 50 selected studies to ensure consistency, completeness, and suitability for structured synthesis. The form was created to capture all information necessary to compare studies across cryptocurrency on-chain and market indicators, methodological approaches, and reported outcomes from a data analytics perspective. All team members participated in the extraction process using the same predefined schema. The form was first tested on a small subset of papers to confirm that every required attribute could be consistently captured before proceeding with full extraction. Since all selected papers were fully accessible and clearly reported their experimental setup and findings, no missing data issues were encountered.

4 Results

4.1 Paper Elimination Summary

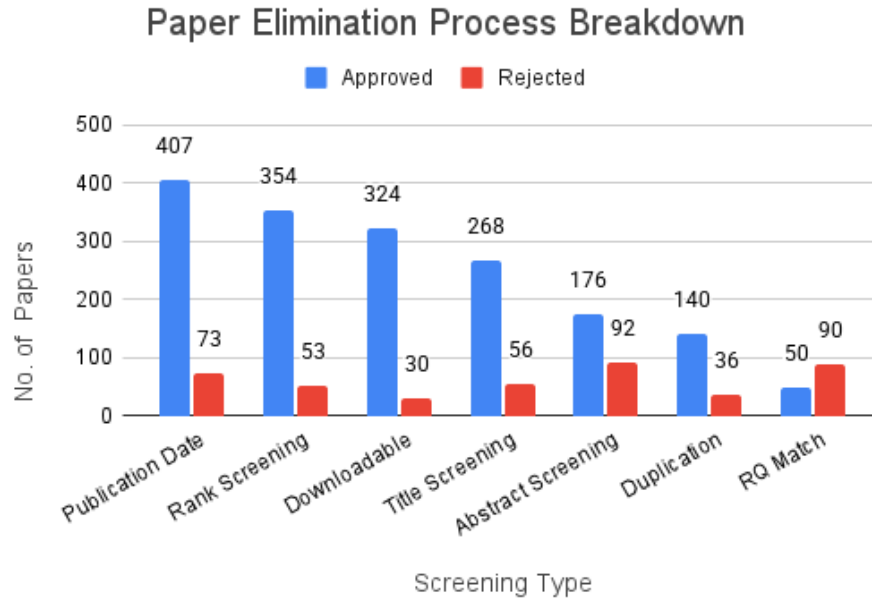


Fig. 2: Paper Elimination Process Breakdown.

This diagram shows the multi-stage screening process used to select 50 relevant papers for the research, with specific screening criteria.

4.1.1 Publication Trends Over the Years

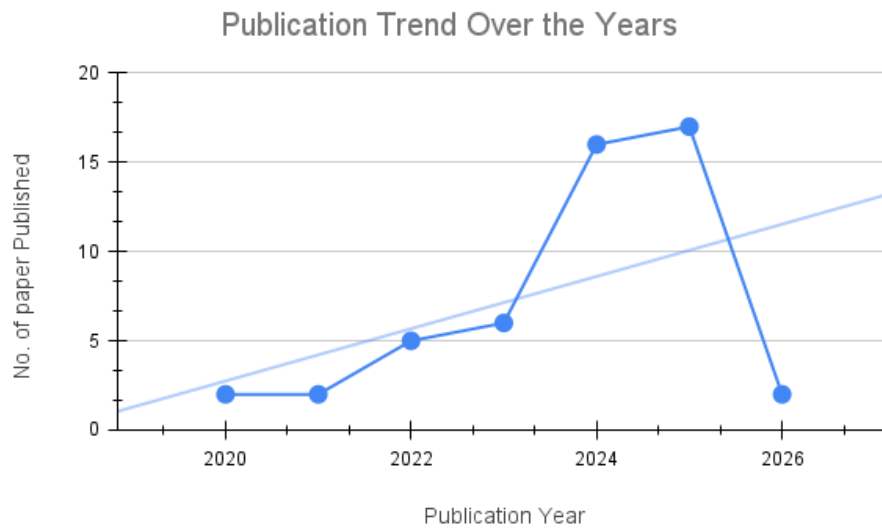


Fig. 3: Publication Trend Over the Years.

The line graph shows the number of publications increasing from 2020 to the projected year 2026, indicating rapid growth in research output.

4.2 Breakthrough Findings: Key Outcomes from Selected Studies

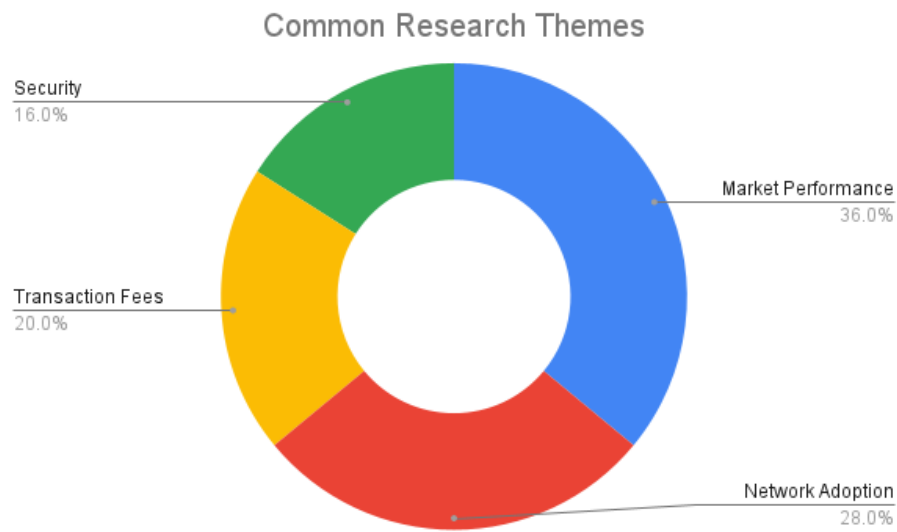


Fig. 4: Common Research Themes.

This chart shows the distribution of key research themes, with market performance being the most studied area at 36 percent. Network adoption follows at 28 percent, transaction fees at 20 percent, and security at 16 percent.

4.2.1 Distribution of Papers by Database

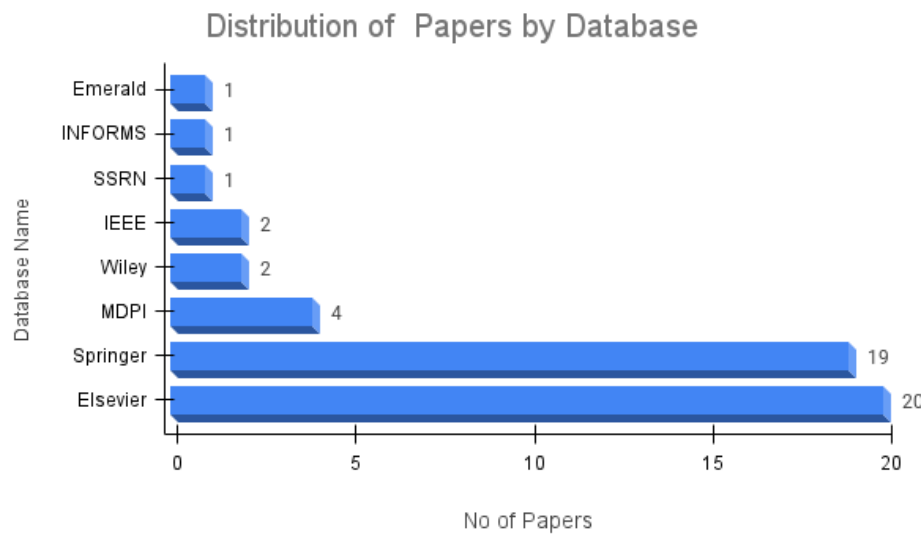


Fig. 5: Distribution of Research Papers by Database.

This graph shows different publishers for 50 papers, where Elsevier and Springer are the dominant sources, together accounting for 39 of the papers, while publishers like Emerald and INFORMS are rarely represented.

4.2.2 Geographical Distribution of Research Publications

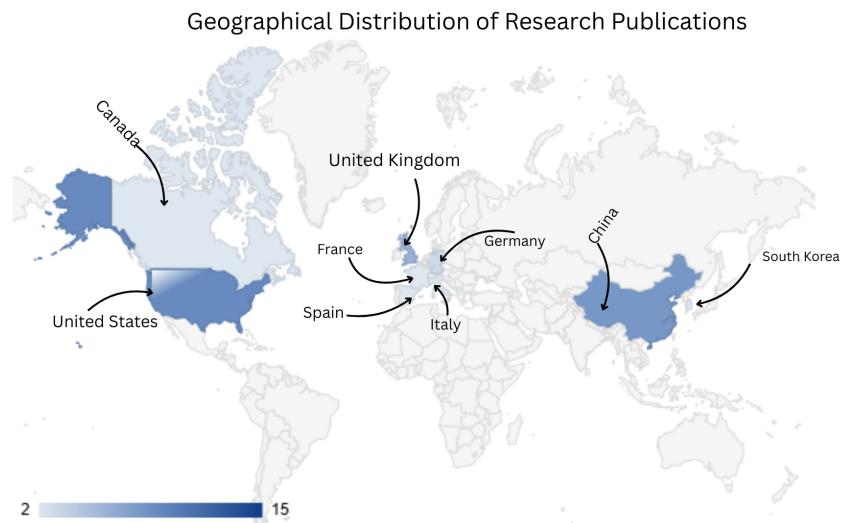


Fig. 6: Geographical Distribution of Research Publications.

This chart shows the geographic distribution of research contributions, with the United States (9 papers) and China (8 papers) as the leading countries. The United Kingdom follows with 6 papers, while a long tail of 15 additional countries/regions each contributed a single paper, indicating a broad but thin global interest.

4.2.3 Research Publications by Focus Area

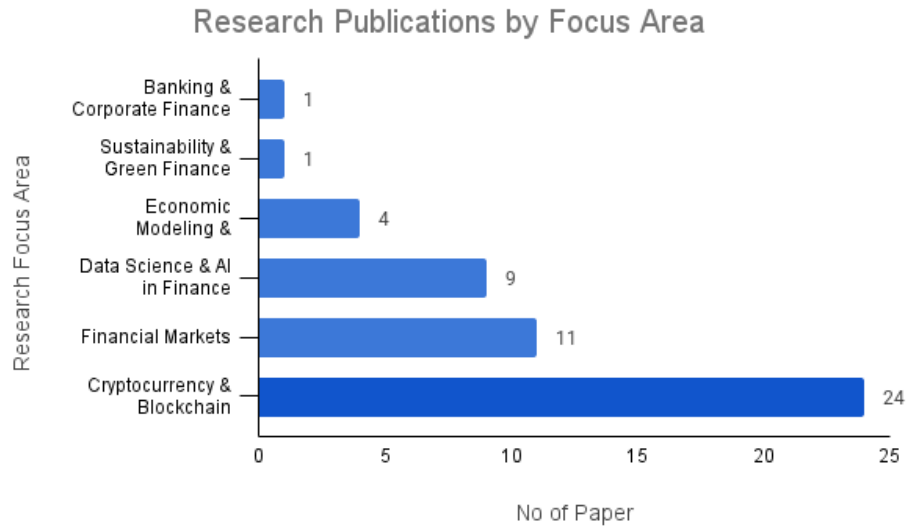


Fig. 7: Research Publications by Focus Area.

This chart ranks the primary focus areas of the research papers, showing cryptocurrency and blockchain as the dominant field with 24 publications. Financial Markets (11) and Data Science and AI in Finance (9) are also significant, while Financial Technology and Behavioral Finance show no dedicated publications in this dataset, indicating current research gaps.

4.3 Methodological Approaches: Research Strategies and Techniques

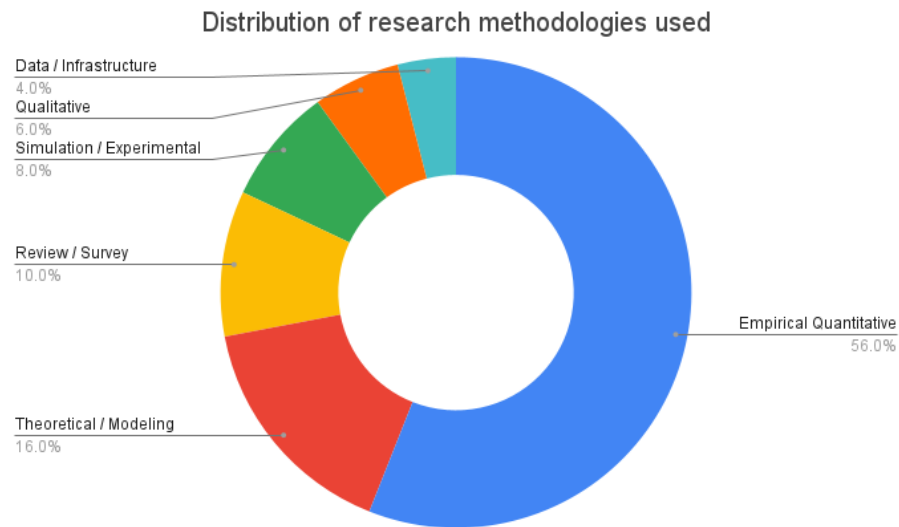


Fig. 8: Distribution of Research Methodologies Used.

This shown chart presents the methodological breakdown of research in percentages, with empirical quantitative being the dominant approach at 56 percent of the papers. The remaining methodologies are less common, including theoretical/modeling (16 percent), review/survey (10 percent), and simulation/experimental (8 percent), highlighting the field's strong emphasis on data-driven analysis.

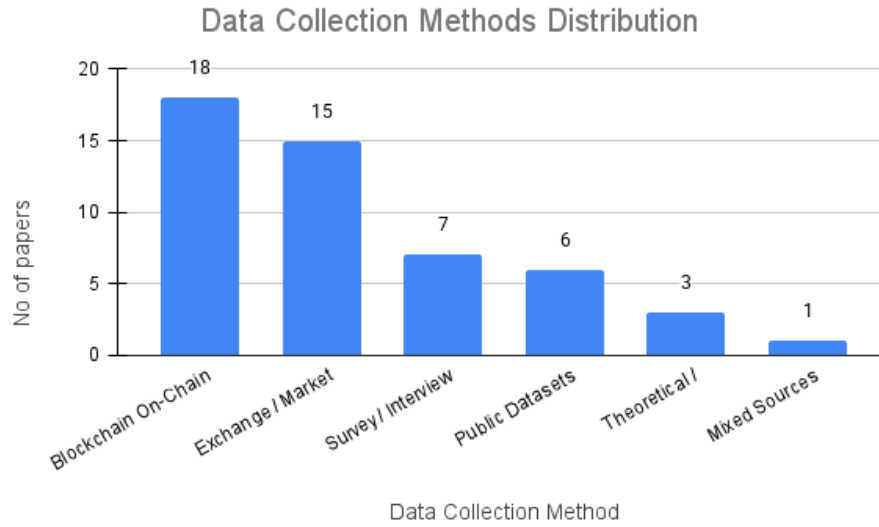


Fig. 9: Data Collection Methods Distribution.

This chart shows that blockchain on-chain data is the most popular collection method (used in 18 papers), while methods like surveys or mixed sources are used far less frequently.

4.4 Cutting-Edge Technologies

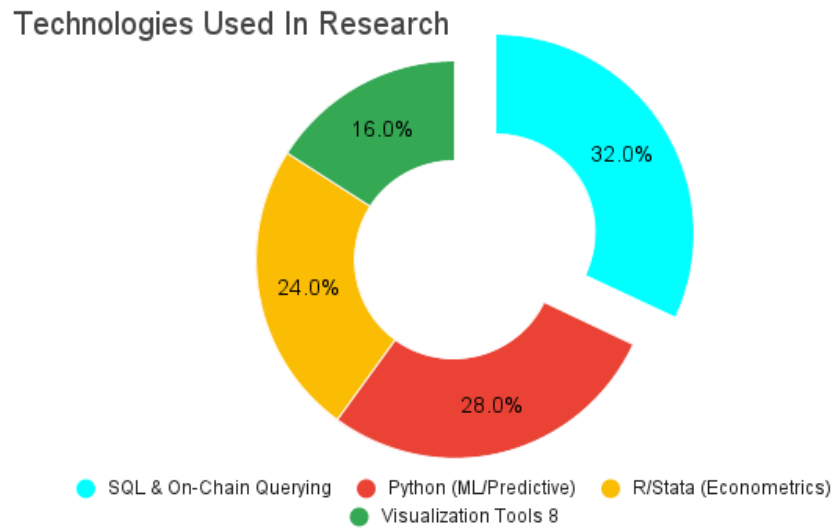


Fig. 10: Technologies Used in Research.

This chart illustrates the primary technologies employed in research, with SQL and On-Chain Querying being the most used at 32 percent, followed by Python (ML/Predictive) at 28 percent, R/Stata (Econometrics) at 24 percent, and Visualization Tools at 16 percent.

4.5 Distributions of Research Papers by Target Audience

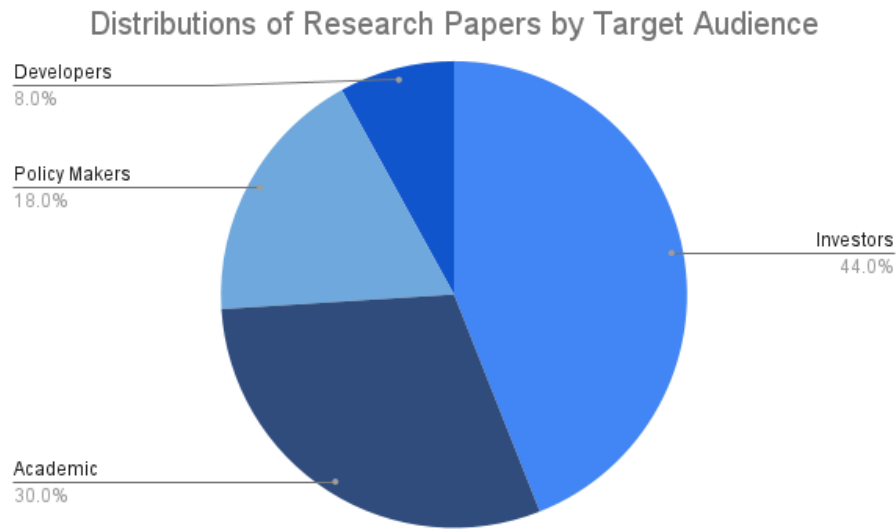


Fig. 11: Distribution of Research Papers by Target Audience.

The audience for these papers is segmented, with investors as the largest group (44 percent), followed by academics (30 percent), policymakers (18 percent), and developers (8 percent).

4.5.1 Barriers to Implementation: Challenges and Limitations

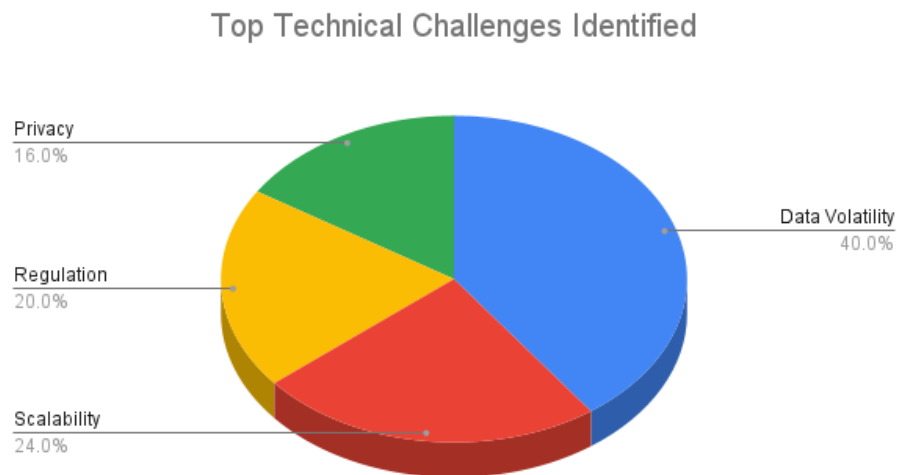


Fig. 12: Top Technical Challenges Identified.

This chart highlights the primary technical hurdles in the field, with data volatility being the most significant challenge at 40 percent. Scalability follows at 24 percent, regulation at 20 percent, and privacy concerns at 16 percent.

4.5.2 Future Prospects: Research Gaps and Emerging Directions

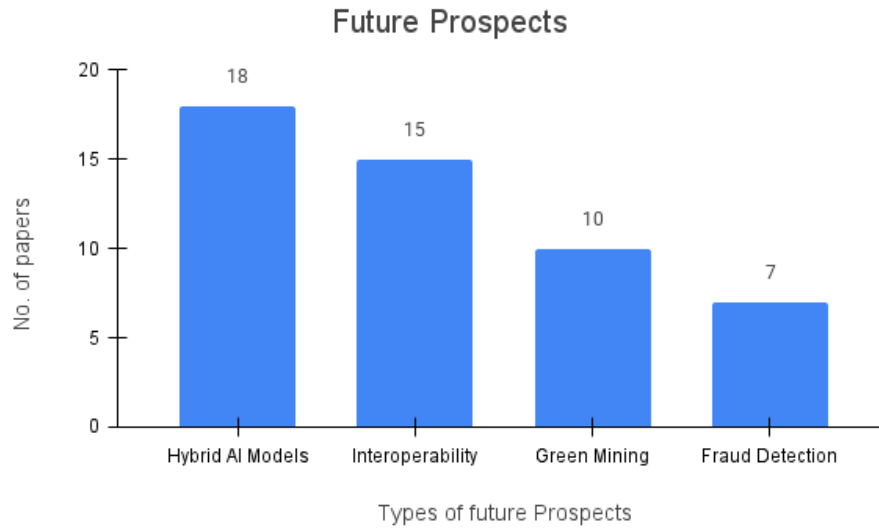


Fig. 13: Future Research Directions Proposed in the Papers.

The distribution of future research priorities across the literature reveals key trends. The development of hybrid AI Models (18 papers) is the most cited direction, followed by Interoperability (15 papers), Green Mining (10 papers), and enhanced Fraud Detection (7 papers). This progression indicates the field's maturation from foundational analysis toward solving complex, systemic challenges in scalability, sustainability, and security.

4.6 Impact on Society: Practical and Social Implications

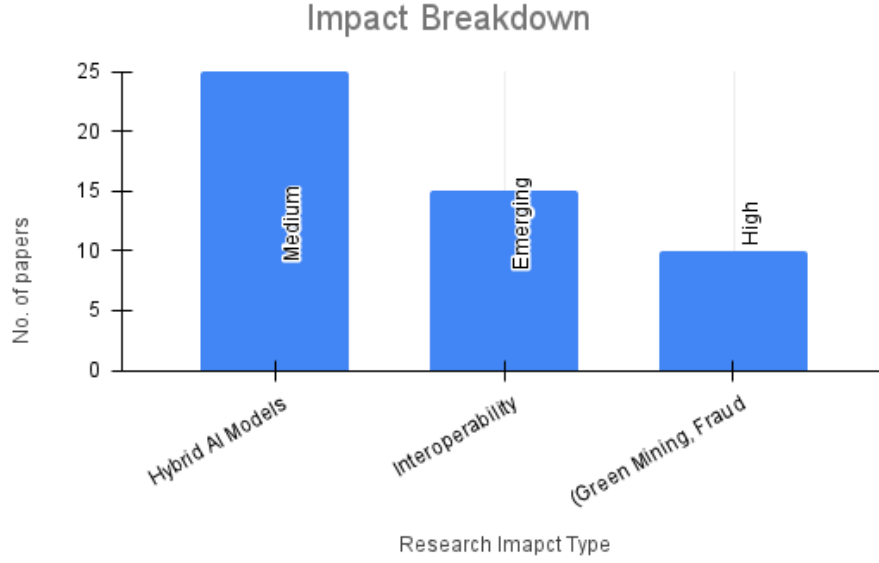


Fig. 14: Breakdown of Societal Impact Identified in the Research.

This chart categorizes key research topics by their perceived impact level and the number of associated papers. Hybrid AI models have the highest volume (25 papers) but are rated as having a medium impact, whereas green mining and fraud detection have fewer papers (10) but are categorized as high impact. Interoperability, with 15 papers, is labeled as an emerging focus area.

4.7 Contribution and Research Questions Matching

This section presents a synthesis of the contributions made by the 50 studies included in this systematic review, highlighting how these contributions map onto the predefined research questions (RQs). A structured coding scheme was applied to classify each study according to its contribution type, including methodological advances, dataset/benchmark creation, evaluation frameworks, application/deployment, and interpretability/calibration. For each RQ, studies were scored as fully answered, partially answered, or not answered. A study was considered to fully answer an RQ if it provided direct empirical evidence, modeling results, or applied analysis addressing the question. Partial answers were assigned when studies provided indirect insights, proxies, or preliminary findings, whereas studies not addressing the RQ were labeled as “not answered.”

4.7.1 Overview of Contributions

The included studies collectively contributed to advancing both the methodology and practical understanding of cryptocurrency adoption, network activity, and financial performance. Five thematic blocks of contributions were identified:

1. **Methodological Advances:** These contributions primarily focused on forecasting cryptocurrency price returns, volatility, and trends using econometric modeling, machine learning, deep learning, and graph neural networks. Typical metrics included prediction accuracy, RMSE, MAE, Sharpe ratios, and realized volatility measures. Representative studies include *Forecasting Bitcoin returns: Econometric time series analysis vs. machine learning*, *Deep learning for Bitcoin price direction prediction*, and *Forecasting cryptocurrency volatility: a novel framework based on the evolving multiscale graph neural network*. The maturity level of these contributions ranged from exploratory to partially validated. Methodological advances directly support RQ2 (effect of user adoption on price returns and volatility) and RQ5 (on-chain metrics predicting weekly returns and volatility).
2. **Dataset and Benchmark Contributions:** Several studies provided comprehensive on-chain datasets, temporal entity–entity transaction graphs, and macroeconomic or liquidity datasets, which serve as a foundation for reproducible analysis. Metrics included wallet counts, active addresses, transaction counts, and network value metrics. Key examples are *Bitcoin research with a transaction graph dataset*, *The Temporal Graph Dataset of Bitcoin Entity–Entity Transactions*, and *Blockchain metrics and indicators in cryptocurrency trading*. These contributions are generally validated and publicly accessible, supporting RQ1 (wallets and active addresses → transaction activity and market growth) and RQ3 (transaction count and volume predicting market trends).
3. **Evaluation Frameworks:** Several studies established model comparison and evaluation frameworks, using techniques such as multiscale backtesting and error variance assessment to benchmark predictive models. Metrics included forecasting accuracy, model stability, and variance of prediction errors. Examples include *Dynamics in Realized Volatility Forecasting: Evaluating GARCH Models and Deep Learning Algorithms Across Parameter Variations* and *Enhancing Cryptocurrency Market Volatility Forecasting with Daily Dynamic Tuning Strategy*. These frameworks contribute partial or full support to RQ2 and RQ5 by validating predictive performance across diverse scenarios.
4. **Application and Deployment Studies:** These studies explored real-world implications of network adoption, transaction fees, and consensus participation, analyzing how these factors influence network growth and long-term value. Metrics included user adoption, transaction fees, hash rates, and NVT ratios. Representative works include *Global Surge: Exploring Cryptocurrency Adoption with Evidence from Spatial Models*, *Blockchain Tokens*, *Price Volatility*, and *Active User Base*,

and *Transaction Fee Economics in the Ethereum Blockchain*. These studies predominantly support RQ1, RQ3, and RQ4, highlighting real-world mechanisms affecting network adoption and financial outcomes.

5. **Interpretability and Calibration:** A smaller subset of studies focused on explainability and feature importance, employing tools such as SHAP values, correlation analyses, and sensitivity testing. Papers such as *Cryptocurrency Returns and Cryptocurrency Uncertainty: A Time-Frequency Analysis* and *Analysis of Cryptocurrency Connectedness Based on Network to Transaction Volume Ratios* provided insights into how on-chain metrics influence market behavior. These contributions support RQ2, RQ3, and RQ5, ensuring model outputs can be interpreted and applied in practice.

Summary

Overall, contributions cluster around methodological innovation and dataset creation, with fewer studies emphasizing interpretability or applied deployment. The strongest evidence is available for predictive modeling of price and volatility, while linkages between adoption, network activity, and long-term value (RQ1 and RQ4) remain under-addressed, highlighting areas for future research.

4.7.2 Research Question Matching Analysis

Using the coding scheme described above, the 50 studies were assessed for coverage of each research question. The aggregated results are presented in Table 1.

Table 1: Research Question Matching Analysis

Research Question	Fully Answered	Partially Answered	Not Answered
RQ1: Do more wallets and active addresses lead to higher transaction activity and market growth?	18	21	11
RQ2: Does user adoption (wallets and active addresses) influence Bitcoin and Ethereum price returns and volatility?	22	19	9
RQ3: How does the relationship between transaction count and volume predict market trends and network usage?	16	26	8
RQ4: How do transaction fee dynamics and consensus participation levels (hash rate or staking) influence network adoption and long-term value in Bitcoin and Ethereum?	12	23	15
RQ5: Which on-chain metrics best predict weekly cryptocurrency returns and volatility?	20	18	12

This analysis shows that RQ2 and RQ5 are most comprehensively addressed, reflecting a strong focus on predictive modeling of cryptocurrency price and volatility. In contrast, RQ1 and RQ4 exhibit lower full coverage, suggesting a gap in studies that empirically link adoption metrics and consensus participation to network growth and long-term value. RQ3 demonstrates moderate coverage, with many studies providing partial insights through proxy measures such as transaction volume and network activity.

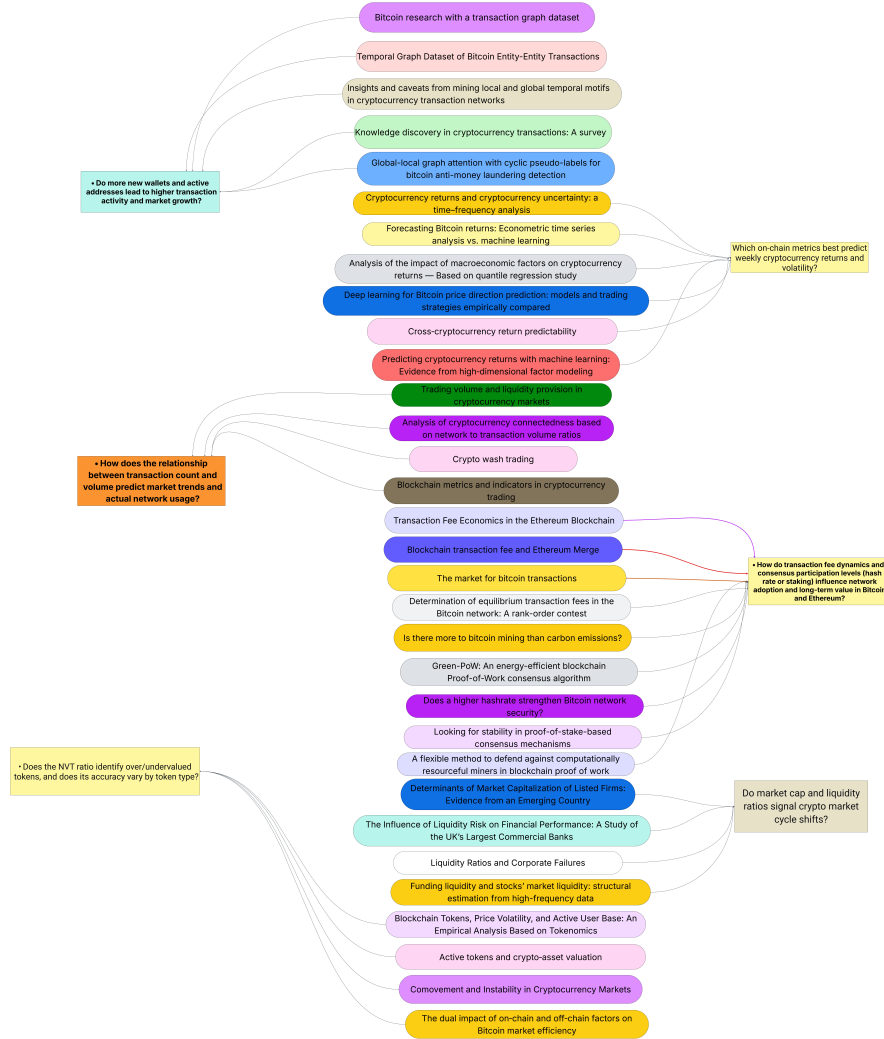


Fig. 15: Research Question Coverage Analysis

5 Comparison and Analysis

Table 2 situates this review within the broader literature on cryptocurrency research. While previous studies have focused either on transactional indicators or behavioral adoption factors, few have integrated these perspectives to examine how user activity impacts measurable market outcomes. Our review addresses this gap by empirically linking on-chain adoption metrics, such as new wallet creation, active addresses, transaction volume, and fees, to financial performance indicators, providing a unified understanding of how network activity translates into returns, volatility, liquidity, and valuation trends

Table 2: Comparison with Other Literature Review Work

Work	Scope	Key Characteristics	Results / Contribution	Our Review
Our Review	Empirically examines how on-chain adoption and network activity metrics such as new wallet creation, daily active addresses, transaction volume, transaction fees, and security participation relate to cryptocurrency financial performance including returns, volatility, liquidity, and valuation indicators such as the NVT ratio.	Follows the PRISMA-2020 systematic literature review framework, prioritizing Q1 journal articles and recent empirical studies. Integrates adoption behavior, network usage, and market outcomes while addressing data-quality concerns such as protocol-induced transaction inflation and wash trading.	Shows that growth in new and active wallet addresses is strongly associated with increased transaction activity and improved network expansion. Demonstrates that on-chain metrics enhance the prediction of returns and volatility when combined with market data.	Identifies robust predictors of cryptocurrency fundamentals, highlights methodological gaps, and provides insights for modeling long-term network sustainability and valuation.
Liu et al. (2021)	Focuses on transaction-based indicators and common risk factors influencing cryptocurrency valuation and price dynamics from a financial economics and asset pricing perspective.	Synthesizes empirical findings on transaction analytics, valuation signals, and spillover risks. Draws parallels with traditional financial markets, with limited attention to adoption-related variables.	Concludes that transaction-based indicators help explain valuation dynamics and risk transmission in cryptocurrency markets.	The role of user adoption in driving financial outcomes remains indirect and insufficiently explored.
Alrehaili et al. (2024)	Examines behavioral, social, and institutional factors influencing cryptocurrency adoption, including trust, perceived risk, regulation, and financial literacy.	Conducts a systematic review of survey- and questionnaire-based studies from social science literature without using on-chain or market data.	Provides comprehensive insights into psychological and socio-economic drivers of adoption.	Does not connect adoption drivers with measurable network activity or cryptocurrency financial performance.

6 Discussion

The findings from the 50 reviewed studies highlight the evolving role of on-chain adoption metrics, network activity indicators, and financial variables in understanding cryptocurrency markets. Overall, research has moved from isolated, descriptive analyses toward integrated modeling frameworks that treat blockchain data as economically meaningful signals rather than raw network statistics. This section synthesizes the key insights, discusses practical implications, and highlights methodological and contextual limitations of the evidence base.

6.1 Key Findings and Implications

This systematic literature review examined how on-chain adoption metrics, network activity indicators, and financial variables interact to explain cryptocurrency market behavior. The synthesized evidence demonstrates a clear shift from descriptive, single-metric analyses toward integrated modeling frameworks that treat blockchain data as economically interpretable signals rather than raw network statistics. Across the literature, on-chain indicators—particularly active addresses, transaction volume, and network value ratios—consistently function as reliable leading indicators of short-term price movements and market volatility. Analytical models that integrate adoption, activity, and financial variables outperform traditional market-only economic approaches, reinforcing the view that cryptocurrency markets operate through complex, non-linear feedback mechanisms linking user behavior, network usage, and price dynamics. However, the relationship between network growth and long-term economic value remains weakly established. While adoption metrics reliably predict transaction activity and short-term market fluctuations, their association with sustained valuation, network security, and long-term ecosystem stability is inconsistent. This suggests that adoption indicators should not be interpreted as standalone proxies for value, but rather as components within broader economic and structural valuation frameworks. A central insight emerging from the literature is that the analytical usefulness of blockchain data depends critically on its economic transformation and adjustment. Metrics that correct for inactive holdings, protocol artifacts, and structural network effects provide substantially more stable and interpretable signals than raw on-chain statistics, highlighting the importance of preprocessing and economic signal extraction.

Implications for Practice and Systems

The findings support the operational integration of cleaned on-chain indicators into analytical and monitoring systems:

- **Investment analytics:** Integrated models combining adoption, activity, and liquidity indicators can enhance forecasting of short-term returns and volatility.
- **Risk monitoring systems:** Indicators such as abnormal transaction activity, active address fluctuations, and fee dynamics can serve as early-warning signals of market instability.
- **Network health assessment:** Ecosystem stakeholders can monitor participation, fee sustainability, and usage intensity as operational measures of network vitality.

- **Policy and market oversight:** Standardized on-chain indicators—such as large-wallet concentration and transaction intensity patterns—can complement traditional financial surveillance tools for assessing market integrity and systemic risk.

Summary: Overall, the findings underscore the need for causal modeling approaches, standardized multi-chain datasets, and adaptive analytical frameworks capable of capturing structural changes in evolving blockchain systems.

6.2 Challenges and Limitations

Limitations of the Evidence Base //

- **Data quality and protocol noise:** On-chain datasets may include wash trading, automated transactions, and protocol-generated activity that distort measures of adoption and network usage.
- **Methodological heterogeneity:** Variations in modeling approaches, time horizons, and metric definitions limit direct comparability across studies.
- **Asset concentration bias:** The dominance of Bitcoin and Ethereum constrains the generalizability of findings to other blockchain ecosystems.
- **Limited causal inference:** Most studies establish predictive associations rather than causal mechanisms linking adoption, activity, and financial performance.
- **Limited real-world validation:** Few studies assess model performance in institutional, regulatory, or operational deployment contexts.

Limitations of the Review Process //

- **Search scope constraints:** Relevant gray literature and recent preprints may not have been fully captured despite multi-database searching.
- **Screening subjectivity:** Study inclusion decisions involve unavoidable reviewer judgment during eligibility assessment.
- **Temporal restriction:** Rapid technological and regulatory developments after 2024 may not be fully reflected.
- **Language restriction:** Inclusion of English-language publications only may introduce geographic and publication bias.

Impact of These Limitations: These constraints limit generalizability across assets and market regimes and restrict causal interpretation. Accordingly, the findings should be interpreted as a synthesis of current empirical evidence rather than definitive economic conclusions. Future research should prioritize standardized datasets, causal inference methods, multi-chain analysis, and longitudinal evaluation.

7 Conclusion

This systematic literature review consolidates fragmented empirical evidence on the measurement and visualization of cryptocurrency adoption, network activity, and financial performance. By synthesizing findings from 50 high-quality studies, several clear insights emerge. First, adoption metrics such as new wallet creation and active

addresses consistently correlate with transaction activity and network growth, though data-quality issues, like inflated volumes from protocol artifacts, require careful cleansing. Second, network activity indicators, particularly transaction counts, fees, and liquidity measures, provide predictive power for short-term returns and volatility, with machine learning and deep learning models outperforming traditional econometric approaches in capturing non-linear market dynamics. Third, the NVT ratio remains a widely used valuation tool, but its reliability improves when adjusted for dormancy and interpreted alongside adoption and transaction trends rather than in isolation. Fourth, consensus participation metrics, including hash rate and staking activity, are critical for understanding network security and long-term sustainability, though empirical evidence linking them directly to fundamental value remains limited. Comparative analysis across studies highlights methodological advances in forecasting, dataset creation, and evaluation frameworks, yet also underscores persistent gaps. Most research remains concentrated on Bitcoin and Ethereum, limiting generalizability across diverse blockchain ecosystems. Causal inference is weak due to reliance on observational data, and market distortions such as wash trading are not consistently addressed. Moreover, while predictive modeling of returns and volatility is well developed, the literature provides less clarity on how adoption and consensus mechanisms translate into enduring fundamental value. Overall, this review demonstrates that on-chain metrics, when rigorously processed and integrated with market data, serve as robust leading indicators for short-term market performance. However, their explanatory power for long-term valuation is still underdeveloped. Future research should prioritize standardized multi-chain datasets, causal methodologies, and interpretability frameworks to bridge the gap between predictive analytics and fundamental analysis. By doing so, scholars and practitioners can advance the reliability of cryptocurrency analytics, enhance risk monitoring, and inform both investment strategies and regulatory oversight.

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